

- (c) Occurrence of ice cap and frozen water  
(d) Occurrence of ozone

**I.A.S. (Pre) 1997**

**Ans. (c)**

In the beginning, Mars was treated equivalent to Earth because of occurrence of ice caps and water ice on Mars. In Mars atmosphere, it has 95.1% carbon dioxide, 2.59% nitrogen, 1.94 % argon, 0.16% oxygen and 210 ppm water etc. Mars polar ice caps and water ice is most relevant for the presence of life on Mars. The average temperature of Mars (– 65°C) is quite similar to Earth (-85°C). Mars rotates on its axis approx. 24.6 hours and revolves around the sun approx. 686.98 days.

**4. Which one of the following may have alien life because of a very conducive environment to life?**

- (a) Jupiter  
(b) Mars  
(c) Europa-The Jupiter's Moon  
(d) Moon-The Earth's Moon

**U.P.P.C.S. (Pre) 2000**

**Ans. (b & c)**

Europa is one of the moons of Jupiter. The Voyager1 spacecraft was launched by NASA to research whether life is possible on it. Europa is frozen and covered with a layer of ice. Its surface seems like the frozen ocean on Earth. The depth of this frozen ocean is approx. 60-150 kilometers. If it is true, then Europa will be a place in the solar system besides Earth, where such an amount of water is available.

**5. Phoenix Mars Lander landed on the surface of Mars on :**

- (a) 27 June, 2008                      (b) 27 May, 2008  
(c) 26 June, 2008                      (d) 26 May, 2008

**Chhattisgarh P.C.S. (Pre) 2008**

**Ans. (d)**

Phoenix Mars Lander was launched on 4<sup>th</sup> August, 2007 and landed on 25 May, 2008 at the North Pole on Mars. On 2 November, 2008 the mission officially ended. Phoenix had given the information about the presence of ice on Mars. Note : Since the exact date is not given in the options the, closest date 26 May, 2008 will be the answer.

## **viii. The Jupiter**

\*Jupiter is the fifth planet from the Sun and the largest in the solar system. Its diameter is 142984 km. The average density of Jupiter is 1326 kg/m<sup>3</sup>. \*The descending order of

planets in the solar system as per their diameter is as follows – Jupiter (142984 km), Saturn (120536 km), , Uranus (51118 km), Neptune (49528 km), Earth (12756km), Venus (12104 km), Mars (6792 km), Mercury (4879 km). \*It takes Jupiter 11.86 years to complete one revolution around the sun. \*As of Feb. 2024, 95 satellites of Jupiter have been discovered. Among these, the 4 largest satellites of Jupiter are **Io, Europa, Ganymede** and **Callisto**. These four are also known as Galilean Moons. They were first seen by Galileo Galilei in January 1610 and recognized by him as satellites of Jupiter. \***Ganymede** is the largest and most massive moon of Jupiter and in the solar system. Jupiter is also known as miniature solar system. Voyager-1 spacecraft was sent to explore the Jovian system in 1977. It explored Europa (one of the four Galilean moons). It provided detailed images of Europa's icy surface. Hence causing many scientists to speculate about the possibility of a liquid ocean similar to those found on Earth. These liquid oceans possibly have a depth of appx. 60-150 km. \*If it is proven then Europa will be the only place in the solar system apart from our planet Earth where water is present in that quantity. \***Jupiter's ring** was first discovered by the **Voyager 1** spacecraft in 1979. These rings are also known as **Jovian Rings**. These rings are mostly made of silicate and dust particles. It is because of the transparency that these rings are difficult to spot however their existence cannot be questioned. Jupiter is primarily composed of hydrogen and helium. Some traces of methane, ammonia hydrogen deuteride, ethane etc. are also found. Jupiter is the fastest spinning planet in the solar system.

**1. The moons of the Planet Jupiter were discovered by –**

- (a) Newton                                      (b) Galileo  
(c) Kepler                                        (d) Copernicus

**Uttarakhand Lower Sub. (Pre) 2010**

**Ans. (b)**

Jupiter is the largest planet of the solar system. The four largest moons (IO, Europa, Callisto, Ganymede) are known as Galilean moons as these were discovered by Galileo Galilei in 1610.

**2. Which is the largest Planet in the Solar System?**

- (a) Jupiter                                      (b) Neptune  
(c) Uranus                                        (d) Saturn

**M.P.P.C.S. (Pre) 1996**

**41<sup>st</sup> B.P.S.C. (Pre) 1996**

**M.P.P.C.S. (Pre) 1990**

**U.P.P.C.S. (Pre) 1990**

**Ans. (a)**